

EECT/CE 7325 Advanced VLSI Design

PROJECT #1: Cell Library for 7nm Due: March 28 (5p)

Project Introduction

For this project you will be creating your standard library of cells which will be used for your final project.

Project Goals

- 1) Layout and verify the following cells:
 - INV
 - NAND2
 - NOR2
 - XOR2
 - MUX2:1
 - OAI21
 - OAI22
 - AOI21
 - AOI22
 - DFF
- 2) All of your cells should be created such that when placed side by side next to each other, no DRC errors occur. All pins must be aligned horizontally as well, with uniform spacing.
- 3) Generate a .lib file for the given set of cells using PrimeLib.
- 4) The height of the pdiff must be 4 fins and the height of the ndiff must be 4 fins as well.

Project Rules & Requirements

- 1) All the cells should have the same height w/ VDD & GND rails aligned horiz.
- 2) The input slew rate is 14 ps (0 to 100% or 100% to 0).
- 3) Assume a 10 fF load capacitance when simulating.

What To Turn In (points are deducted for anything missing)

- 1) A cover page containing all the following information.
 - Name, student number, “EECT 7325” or “CE 7325”, and project title
- 2) Each cell layout with **rulers** showing the dimensions of the cell. Part of grade depends on the clarity of your **report**.
 - Show the distance between your pins
 - Show the height & width of entire cell (height & width are measured from implant (IMP) to implant).
- 3) Also turn in the following for each cell in your library:
 - Simulation data that each cell is correctly functional.
- 4) Please also hand in a layout showing all of your cells lined up in a row with the boundary layers touching to demonstrate that they can be placed next to each other with no problems (yes, that means run DRC).
 - Show distance between pins to verify that they have a uniform pitch.
- 5) An electronic copy of the report must be turned in.

Grading Breakdown

Correct functionality of all cells	50%
Correct pin spacing and cell sizing	20%
Report	30%